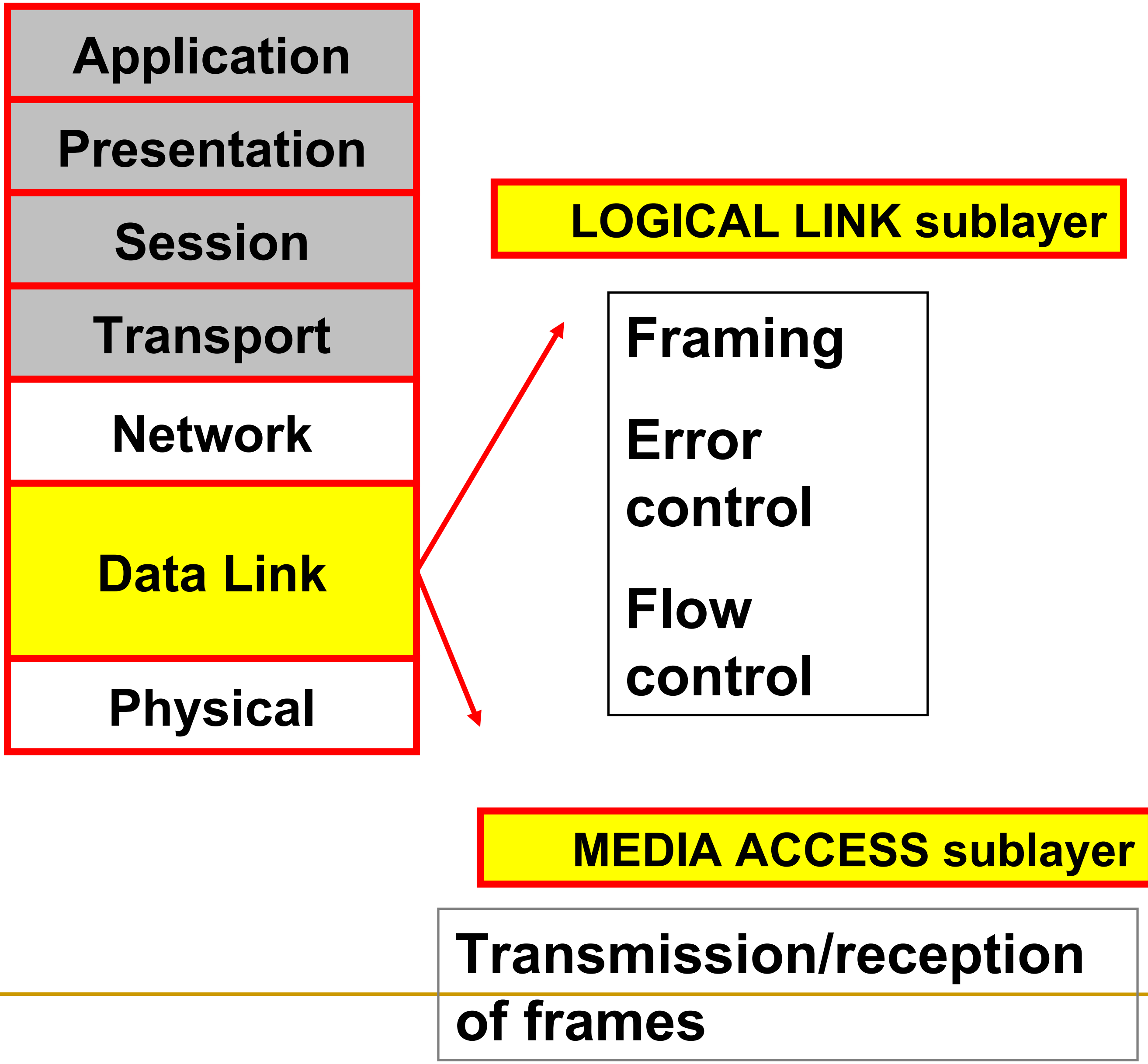
Medium Access Sub Layer
Unit-III

OSI


Application
Presentation
Session
Transport
Network
Data Link
Physical
LOGICAL LINK sublayer
Framing
Error control
Flow control
MEDIA ACCESS sublayer
Transmission/reception of frames

The Medium Access Sub layer

deals with

BROADCAST NETWORKS AND THEIR PROTOCOLS

*Broadcast channels are sometimes referred to as **multi-access channels** or **random access channels.***

Topics

- Introduction
- Channel Allocation problem
- Multiple Access Protocols
- IEEE Standard 802 for LANs
- Wireless LAN
- Bridges & its types.

Introduction

- *Medium Access Control (MAC) sub layer is part of Data Link layer.*
- *In fact, it is the bottom part of DLL (interfacing with the physical layer)*
- *This chapter deals with broadcast networks*

Channel Allocation

- Channel Allocation is a process in which a single channel is divided and allotted to multiple users in order to carry user specific tasks.
- A central controller interrogates each host and allocates channel capacity to those who need it called polling.
- The two main methods for channel allocation are Static Channel Allocation and Dynamic Channel Allocation.

1.Static Channel Allocation.

2. Dynamic Channel Allocation.

The Channel Allocation Problem

- *Static Channel Allocation in LANs and MANs*
- *Dynamic Channel Allocation in LANs and MANs*

Channel Allocation problem

- *Static Channel Allocation in LANs and MANs*
- *FDM & TDM*
- FDM:

1. The traditional way of allocating a single channel, among multiple competing users is Frequency division multiplexing(FDM).
2. If there are N users, the bandwidth is divided into N equal-sized portions, each user being assigned one portion.
3. Here each user has a private frequency band, there is no interference between users.

4. When there is only a small and constant number of users, each of which has heavy load of traffic, FDM is simple and efficient.

5. When number of senders is large and continuously varying, FDM presents few problems.

6. If spectrum is cut up into N regions and <N users are interested in communication, the spectrum is wasted.

7. If more than N users want to communicate, some of them will be denied permission for lack of bandwidth.

8. Assuming that the number of users be held constant at N, dividing the channel into sub-channels is inefficient.

9. The basic problem is, when some users are quiet their bandwidth is simply lost. They are not using it, and no one else is allowed to use it either.

TDM:

- The same arguments that apply to FDM also apply to TDM.
- Each user is statically allocated every N^{th} time slot. If a user does not use the allocated slot, it just lies fallow.
- The same holds if we split up the network physically.

Dynamic Channel Allocation

- The allocation of the channel changes based on the traffic generated by the user.
- If the traffic increases more channels are allocated, otherwise fewer channels are allocated to the users.
- Gives better utilization and lower delay on a channel when the traffic is unpredictable.

Dynamic Channel Allocation in LANs and MANs

- *Station Model.*
- *Single Channel Assumption.*
- *Collision Assumption.*
- *(a) Continuous Time.*
  (b) Slotted Time.
- *(a) Carrier Sense.(LAN)*
  (b) No Carrier Sense.(SATELLITE)

1.Station Model:

- The model consists of N independent Stations, each with a program or user that generates frames for transmission.
  Stations are called Terminals.

2. Single Channel Assumption:

- A single channel is available for all communication.
- All stations can transmit on it and all can receive from it.

3. Collision Assumption:

- If two frames are transmitted simultaneously, they overlap in time and the resulting signal is garbled. This event is called a Collision.
- All stations can detect collisions.
- A collided frame must be transmitted again later.

4a. Continuous Time:

- Frame transmission can begin at any instant.
- There is no master clock dividing time into discrete intervals.

4b.Slotted Time:

- Time is divided into slots. Frame transmission always begins at the start of a slot.
- A slot can have 0,1, or more frames, corresponding to an idle slot, a successful transmission, or a collision respectively.

5a. Carrier Sense:

- Stations can tell if the channel is in use before trying to use it.
- If the channel is sensed as a busy, no station will attempt to use it until it goes idle.

5b. No Carrier Sense:

- Stations cannot sense the channel before trying to use it.
- They just go ahead and transmit. Only later can they determine whether the transmission was successful.

Multiple Access Protocols

- *ALOHA*
- *Carrier Sense Multiple Access Protocols*
- *Collision-Free Protocols*
- *Limited-Contention Protocols*
- *Wavelength Division Multiple Access Protocols*
- *Wireless LAN Protocols*

Medium Access Sub Layer

ALOHA :

- *It is a random access protocol.*
- *It was designed for WLAN but it is also applicable for shared medium.*
- *Multiple stations can transmit the data at the same time can hence lead to collision and data being garbled.*

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Aloha

- *Pure ALOHA (Mr. Norman Abramson in 1970s)*
- *Slotted ALOHA (Mr.Roberts in 1972)*

Pure ALOHA

- *Users transmit whenever they have data to be sent.*

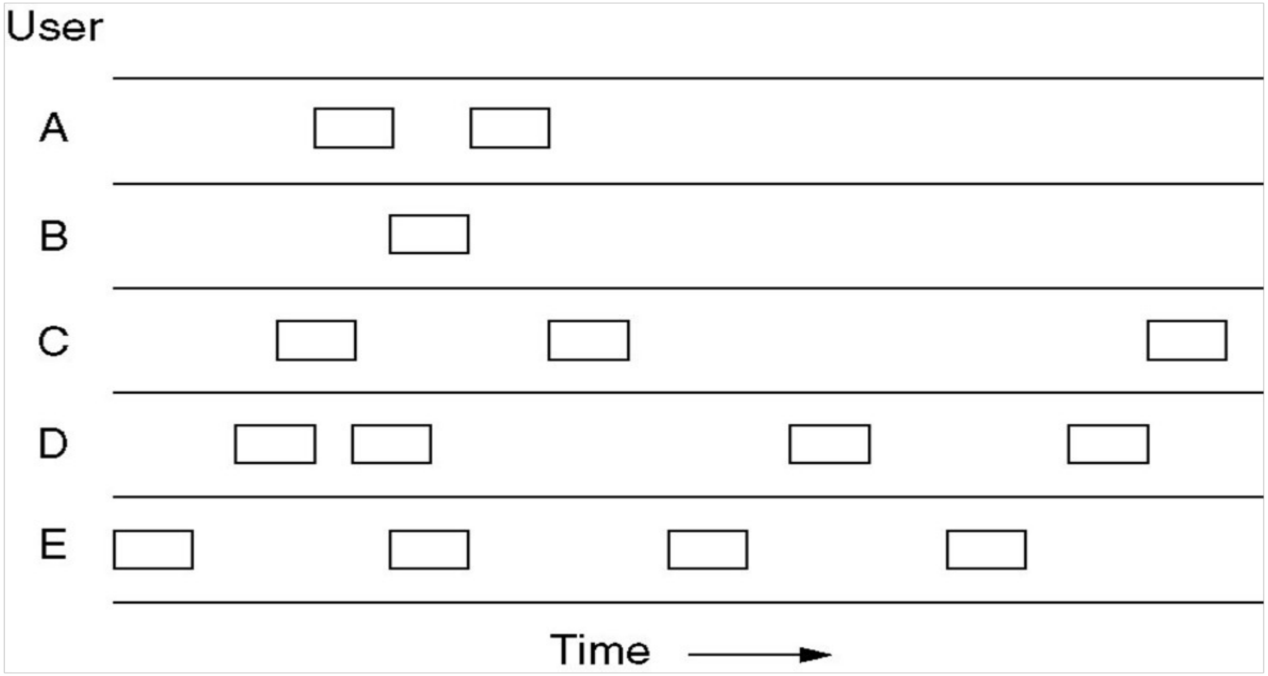


Fig. In pure ALOHA, frames are transmitted at completely arbitrary times

- *Systems in which multiple users share a common channel in a way that can lead to conflicts are widely known as* ***contention*** *system.*
- *The frame size is fixed because the throughput of ALOHA systems is maximized.*
- *We need to resend the frames that have been destroyed during transmission.*
- *A collision involves two or more stations. If all these try to resend their frames after the time-out, the frames will collide again.*
- *Pure ALOHA states that when the time-out period passes, each station waits a random amount of time before resending its frame(T_B).*

- *Pure ALOHA has a second method to prevent congesting the channel with retransmitted frames. After a maximum no.of retransmission attempts K_{max}, a station must give up and try later.*
- *The time-out period is equal to maximum possible round-trip propagation delay, which is twice the amount of the time required to send a frame between the two most widely separated stations($2 * T_p$).*
- *The back-off time TB is a random value that depends on K.*

Figure 12.4 *Procedure for pure ALOHA protocol*

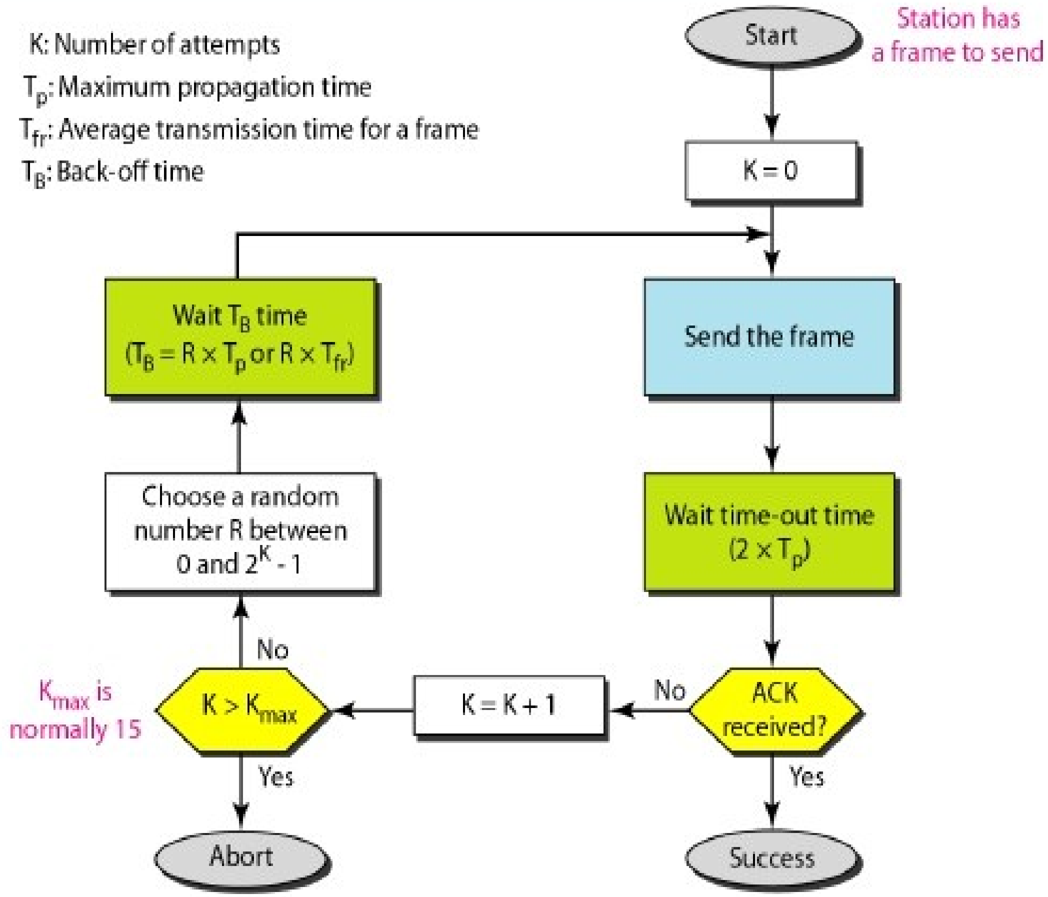
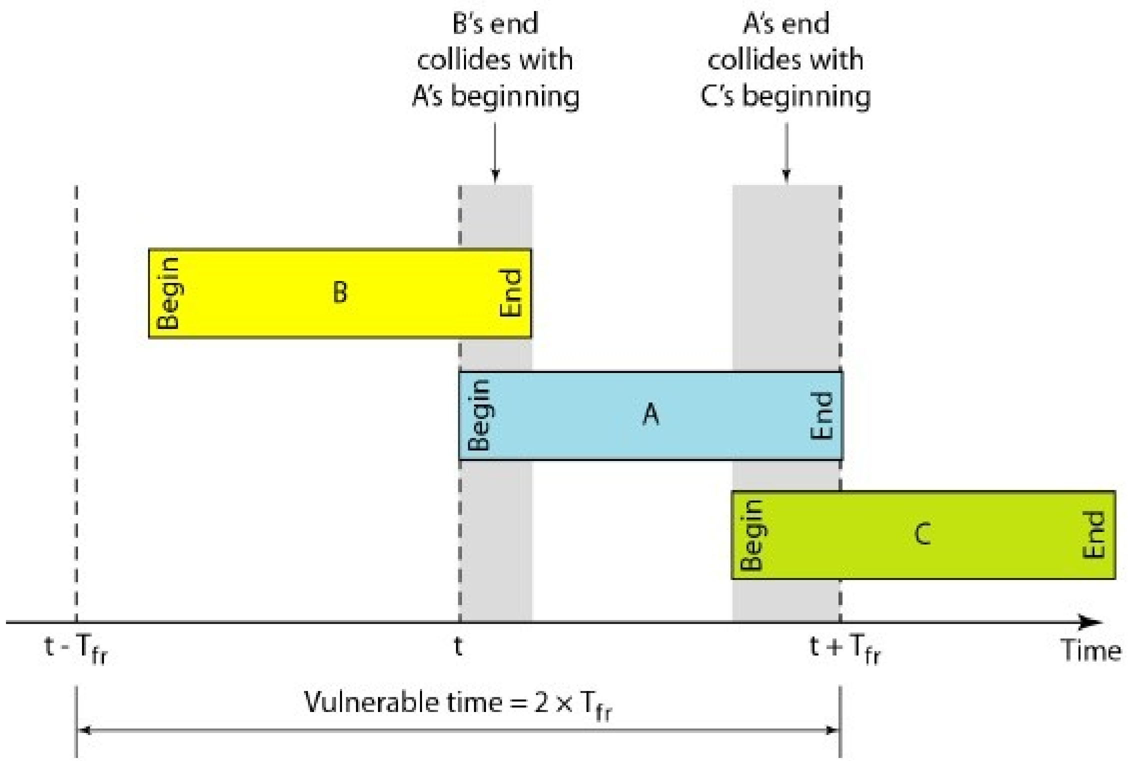


Figure 12.5 *Vulnerable time for pure ALOHA protocol*



Pure ALOHA

- *Whenever two frames try to occupy the channel at the same time, there will be a collision and both will be garbled.*
- *If the first bit of the new frame overlaps with the last bit of a frame almost finished , both frames will be totally destroyed, and both will be retransmitted later.*
- *Throughput for pure ALOHA decreases.*

Pure ALOHA

Disadvantage

- *More number of users share common channels in a way that can lead to conflicts.*
- *More number of collisions occur.*
- *Collision detected: stations waits a random amount of time.*

Pure ALOHA

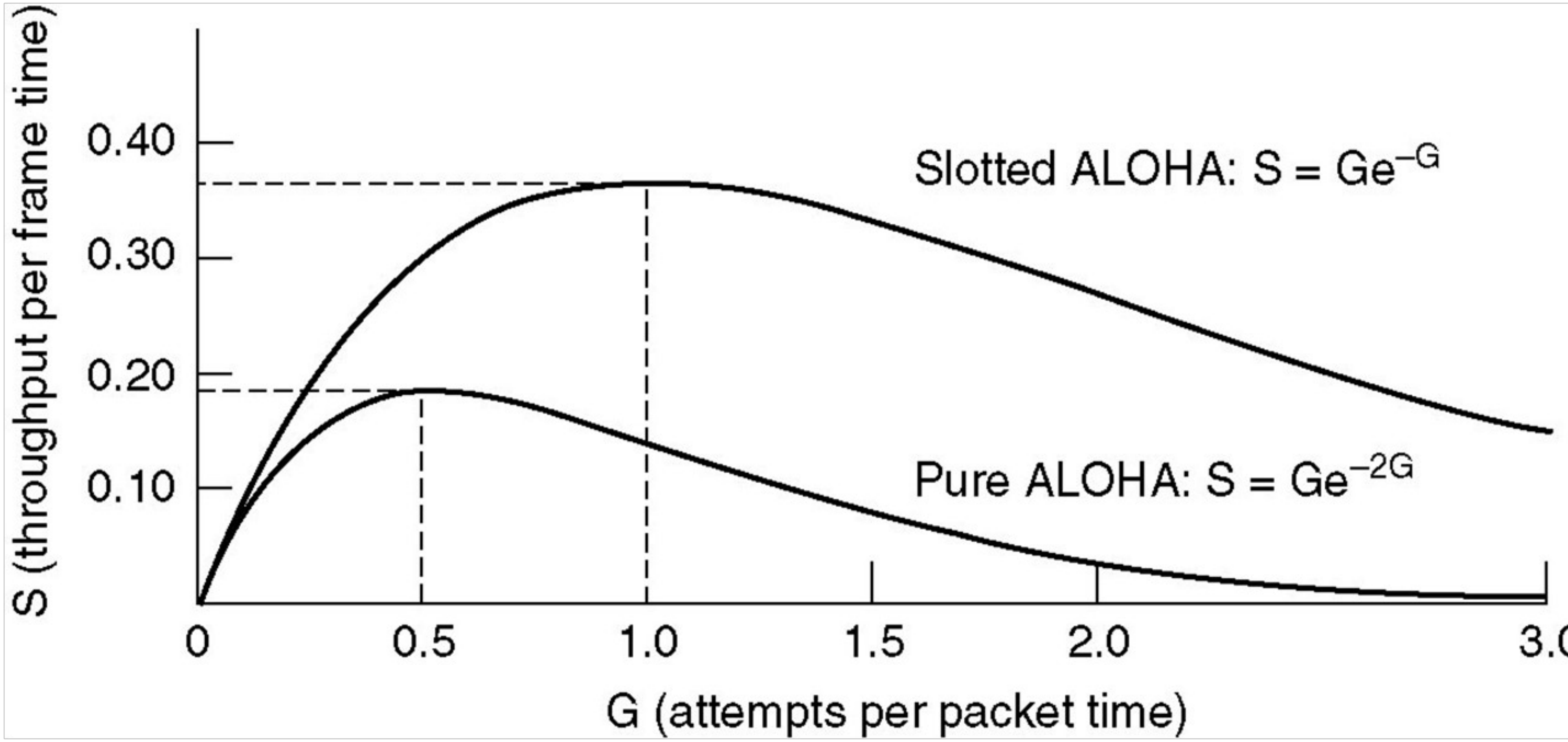
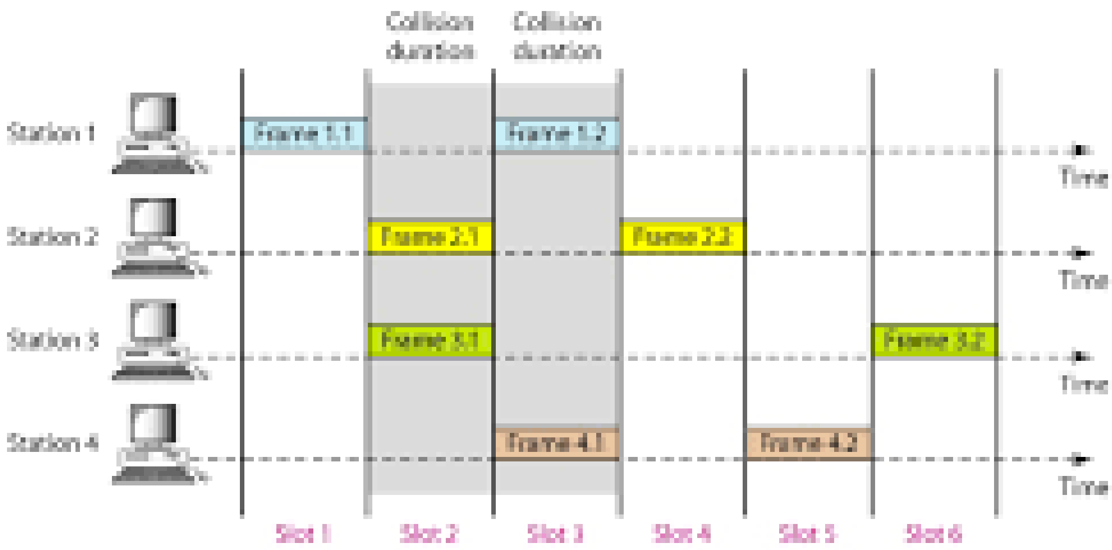


Fig. Throughput versus offered traffic for ALOHA systems

Slotted aloha

- *Slotted ALOHA: Divide the time into discrete intervals(slots) of Tfr s, and force the station to send only at the beginning of the time slot..*
- *Obviously, there may be a special signal needed to synchronize the clocks at all stations.*
- *Because a station is allowed to send only at the beginning of synchronized time slot, if a station misses this, it must wait until the beginning of next time slot.*
- *This means that the station which started at the beginning of this slot has already finished sending its frame.*
- *Ofcourse, there is still the possibility of collision if two stations try to send at the beginning of the same slot.*

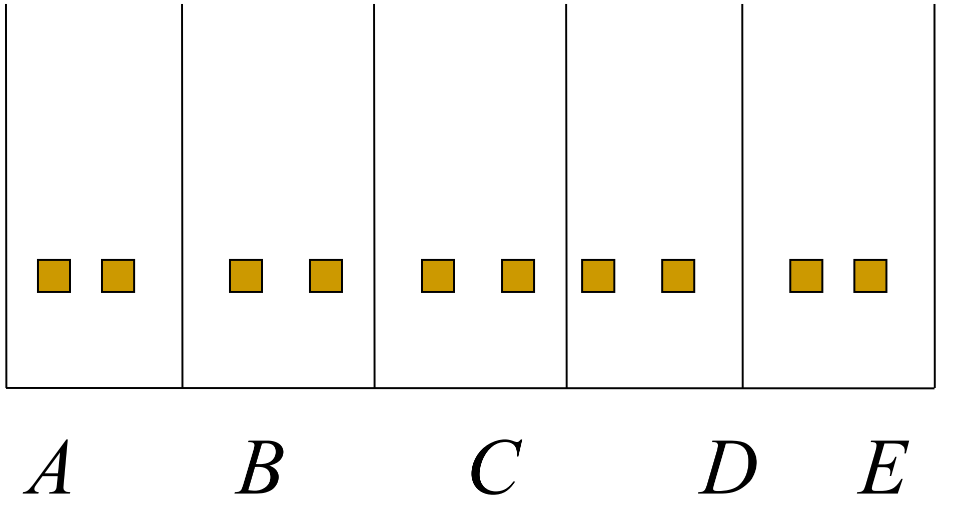
Figure 12.6 *Frames in a slotted ALOHA network*



- *However, the vulnerable time is now reduced to one-half, equal to T_{fr}.*
- *It can be proved that the average number of successful transmission for the slotted ALOHA is $S=G * e^{-G}$*
- *Maximum throughput occurs at G=1, S=1/e or 0.368. This is twice that of pure ALOHA protocol.*
- *In other words, if a frame is generated during one frame transmission time, then 36.8 % of these frames reach their destination successfully. This result is expected because vulnerable time is equal to the frame transmission time.*
- *Therefore, if a station generates only one frame in this vulnerable time(and no other station generates a frame during this time), the frame will be reach its destination successfully.*

Slotted ALOHA

Divides the time into discrete intervals



Disadvantage: collisions
Throughput for slotted ALOHA increases.

Differences between Pure ALOHA and slotted ALOHA

Pure ALOHA	Slotted ALOHA
Any Station can transmit data at any time.	Any station can transmit the data only at the beginning of any time slot.
Vulnerability time = 2 x Transmission time.	Vulnerability time = Transmission time.
The probability of successful transmission in Pure ALOHA is $G \times e^{-2G}$	The probability of successful transmission in Slotted ALOHA is $G \times e^{-G}$
The max probability of successful transmission in pure aloha occurs at $G=½$ and the value is $½ \times e^{-1} = 18.39\%$	The max probability of successful transmission in slotted aloha occurs at $G=1$, the value is $1 \times e^{-1} = 36.79\%$
Time is continuous and not synchronised.	Time is discrete and Synchronised.

CSMA: Carrier Sense Multiple Access

- *Protocols in which stations listen for a carrier (i.e., transmission) and act accordingly are called carrier sense protocols.*

Adv:

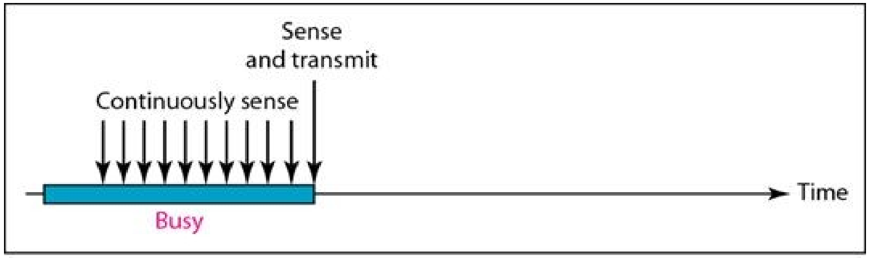
- *To minimize the chance of collision*
- *Increases the performance.*

  - *CSMA principle is "sense before transmit" or "listen before talk".*

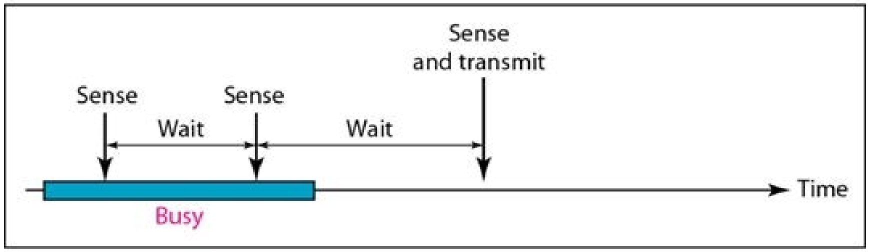
CSMA Methods

- *1-persistent CSMA- constant length packets.*
- *non-persistent CSMA- to sense the channel.*
- *p-persistent CSMA*

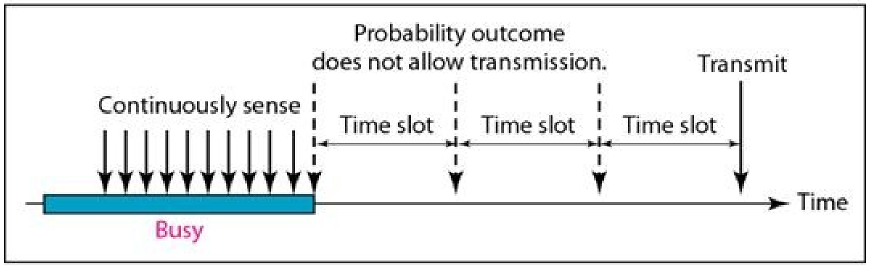
Figure 12.10 *Behavior of three persistence methods*



a. 1-persistent



b. Nonpersistent



c. p-persistent

1-Persistent-after station finds the line idle, send its frame

Nonpersistent-senses the line; idle: sends immediately; not idle: waits random amount of time and senses again

p-Persistent-the channel has time slots with duration equal to or greater than max propagation time

1-persistent

- *When a station has data to send, it first listens to channel to see if any one else is transmitting at that moment.*
- *If the channel is busy, the station continuously senses the channel until it becomes idle.*
- *When the station detects an idle channel, it transmits a frame.*
- *If a collision occurs, the station waits a random amount of time and starts all over again.*
- *The station transmits with a probability of 1 whenever if finds the channel idle.*
- *This method has highest chance of collision because two or more stations may find the line idle and send their frames immediately.*

Non-persistent CSMA

- *A station that has a frame to send it senses the line.*
- *If the line is idle, it sends immediately.*
- *If the line is not idle, it waits a random amount of time and then senses the line again.*
- *This approach reduces the chance of collision because it is unlikely that two or more stations will wait the same amount of time and retry to send simultaneously.*
- *This algorithm should lead to better channel utilization and longer delays than 1-persistant CSMA.*

P-persistent CSMA

- *This method is used if the channel has time slots with a slot duration equal to or greater than the maximum propogation time.*
- *It combines advantages of the other two strategies.*
- *It reduces the chance of collision and improves efficiency.*

P-persistent CSMA

- *In this method, after station finds the line idle it follows these steps:*

1.With probability 'p', the station sends its frame.

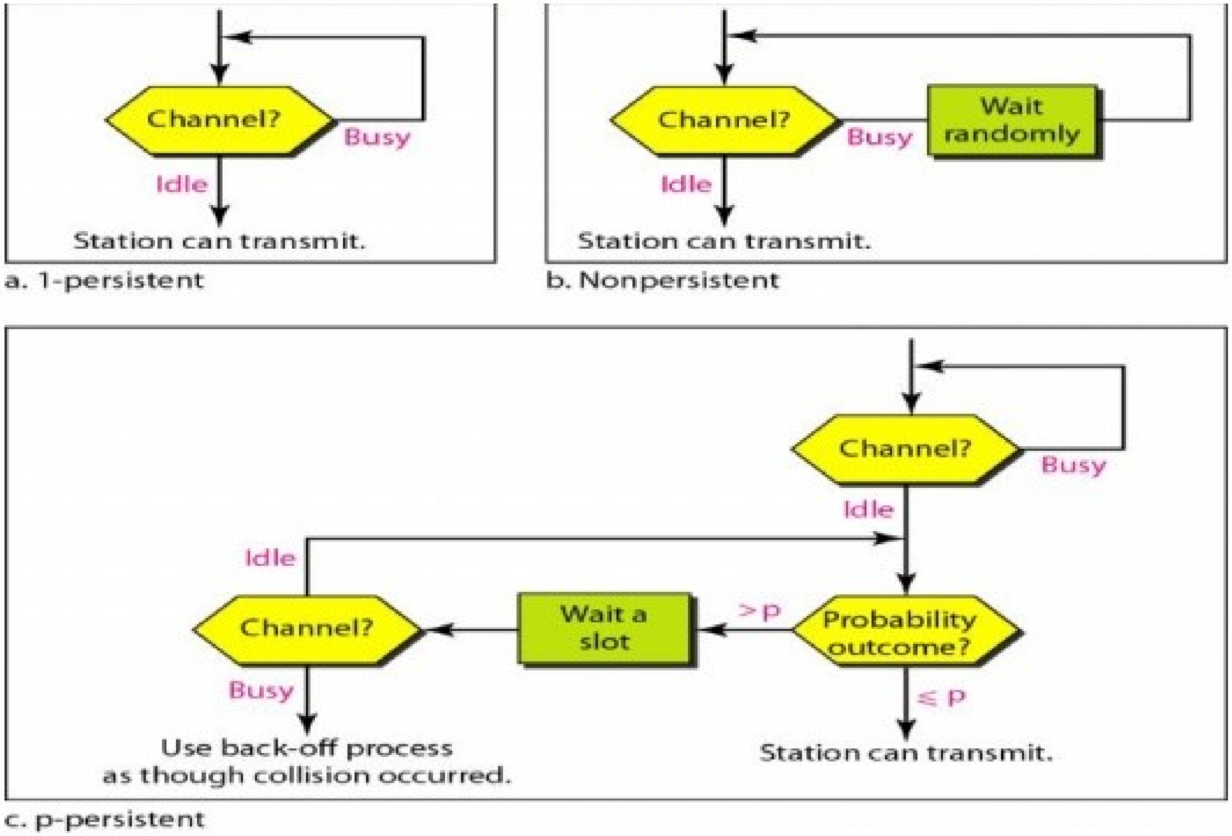
2.With probability q=1-p, the station waits for the beginning of the next time slot and checks the line again.

a. If the line is idle, it goes to step 1.

b. If the line is busy, it acts as though a collision has occurred wait for a random amount of time and starts again.

3.If the station intially senses the channel busy,it waits until the next slot

Figure 2.39 shows the flow diagrams for these methods.



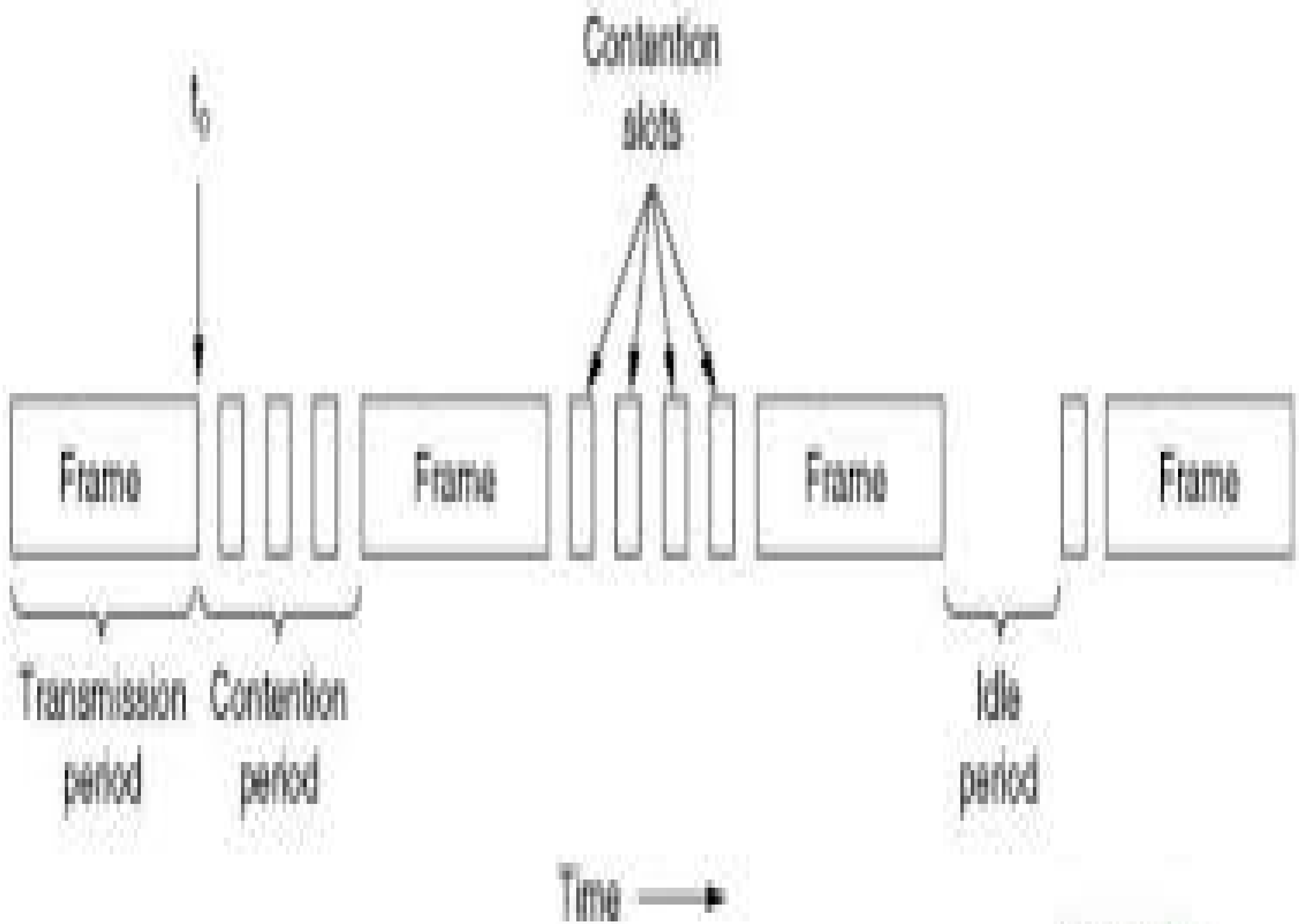
Carrier Sense Multiple Access with Collision Detection(CSMA/CD)

- If two stations sense the channel to be idle and begin transmitting simultaneously, they will both detect the collision immidiately.
- Rather than finish transmitting their frames which are irretrievably garbled anyway.they should abruptly stop transmitting as soon as the collision is detected.
- Quickly terminating damaged frames saves time and bandwidth.

- It is widely used in LANs in the MAC sublayer.
- Access method used by Ethernet: CSMA/CD.

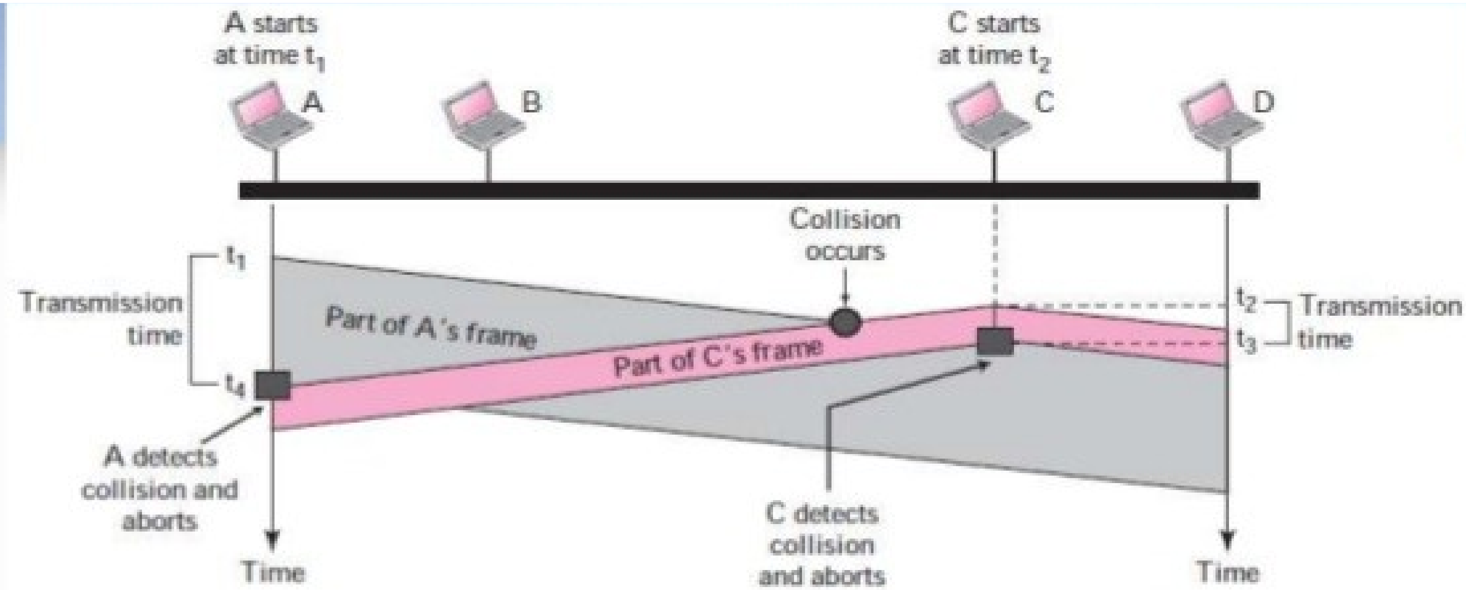
- At the point marked t0, a station has finished transmitting its frame.
- Any other station having a frame to send may now attempt to do so.
- If 2 or more stations decide to transmit simultaneously, there will be a collision.
- Collisions can detected by looking at the power or pulse width of the receiver signal and comparing it to the transmitted signal.

- After a station detects a collision, it aborts its transmission, waits a random period of time and then tries again.
- Therefore, model for CSMA/CD will consist of alternating contention and transmission periods with idle periods occurring when all are quite.



Contention slots
t_0
Frame
Frame
Frame
Frame
Transmission period
Contention period
Idle period
Time

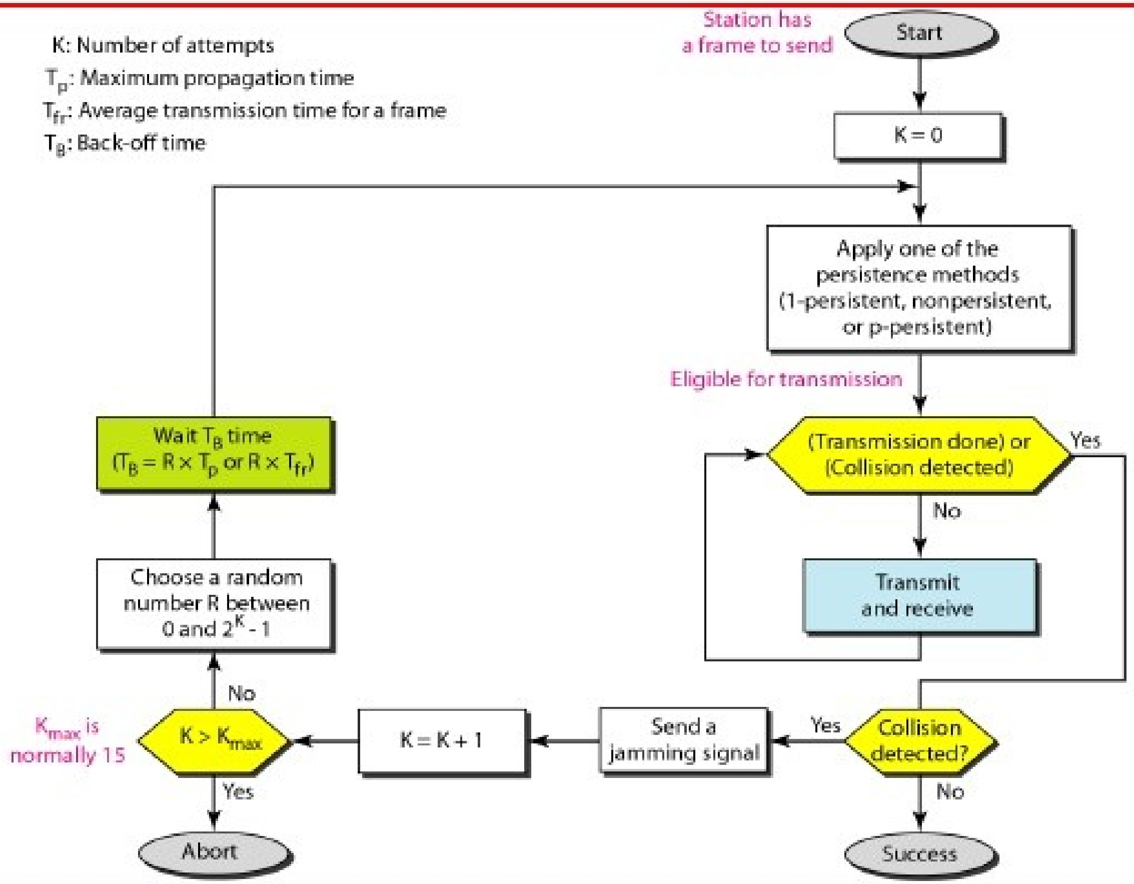
CSMA-CD



Collision of the first bit in CSMA/CD

- A transmits for the duration t4 - t1; C transmits for the duration t3 - t2 → for the protocol to work, the length of any frame divided by the bit rate in this protocol must be more than either of these durations
- Before sending the last bit of the frame, the sending station must detect a collision, if any, and abort the transmission → because, once the entire frame is sent, station does not keep a copy of the frame and does not monitor the line for collision detection

Figure 12.14 *Flow diagram for the CSMA/CD*



CSMA/CD Vs ALOHA

- CSMA/CD is similar to the one for ALOHA protocol, but there are differences which as follows:

1. Addition of persistence methods
2. Frame transmission: In ALOHA, we first transmit the entire frame and then wait for an acknowledgment.

   In CSMA/CD, transmission and collision is a continuous process. We do not send entire frame and then look for a collision. The station transmits and receives continuously and simultaneously.

Contd..

- We use loop to show that transmission is a continuous process.
- We constantly monitor in order to detect one of the two conditions: either transmission is finished or a collision is detected. Either event stops transmission.
- When we come out of loop, if a collision has not detected, it means that the transmission is complete: the entire frame is transmitted. Otherwise, a collision has occurred.

Throughput of CSMA/CD

- The throughput of CSMA/CD is greater than that of pure or slotted ALOHA.
- The maximum throughput occurs at a different value of G and is based on the persistent method and the value of *p* in the *p*-persistent approach.
- For 1-persistent method the maximum throughput is around 50 % when G=1.
- For non-persistent method, the maximum throughput can go up to 90 % when G is between 3 and 8.

Wired LANs: *Ethernet*

1. IEEE Standards
2. Standard Ethernet
3. Fast Ethernet
4. Gigabit Ethernet
5. Ten Gigabit Ethernet.

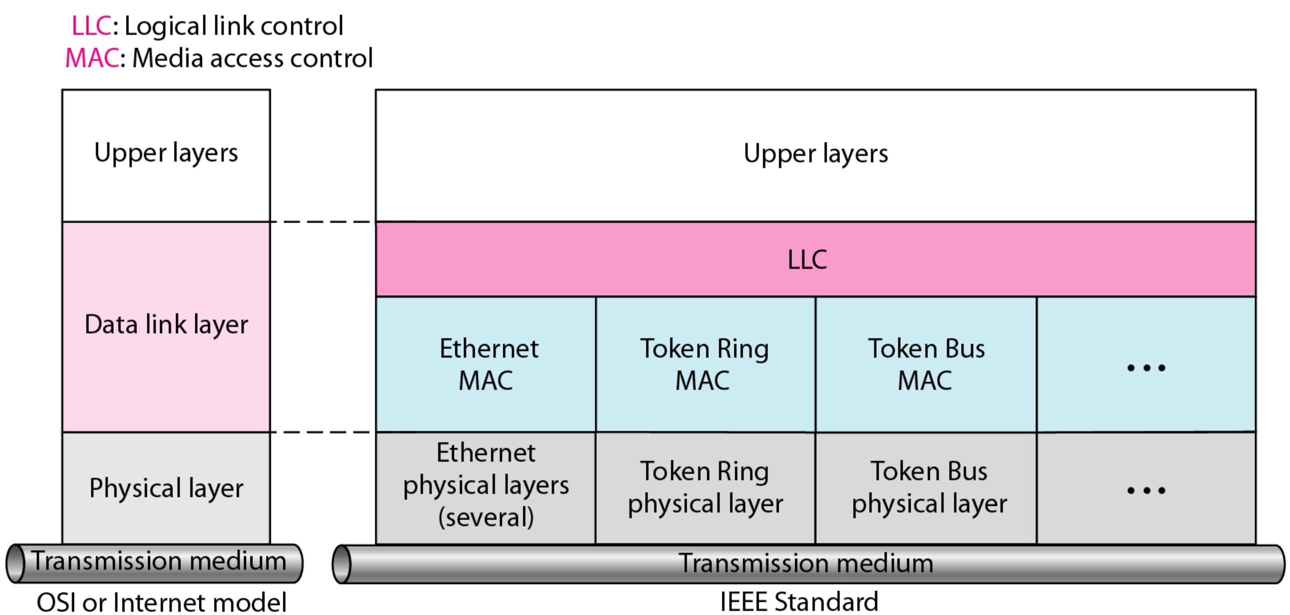
- WHAT IS Ethernet?
- Ethernet is the most common wired standard and technology for connecting devices within a local area network (LAN) or other networks.
- It allows computers, printers, and servers to communicate by transmitting data in packets over cables like twisted-pair or fiber optic.
- It uses unique MAC addresses for device identification.

Ethernet

- It is most widely used wired LAN technologies.
- Operates in the data link layer and physical layer.
- Family of networking technologies that are defines in the IEEE 802.2 and 802.3 standards.
- Supports data bandwidth of 10,100,1000,10,000,40,000,100,000 Mbps.

IEEE Standards

- In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards to enable intercommunication among equipment from a variety of manufacturers. Project 802 is a way of specifying functions of the physical layer and the data link layer of major LAN protocols.



Logical Link Control (LLC)

- Framing: LLC defines a protocol data unit (PDU) that is similar to that of HDLC
- To provide flow and error control for the upper-layer protocols that actually demand these services.
- It provides one single data link control protocol for all IEEE LANs

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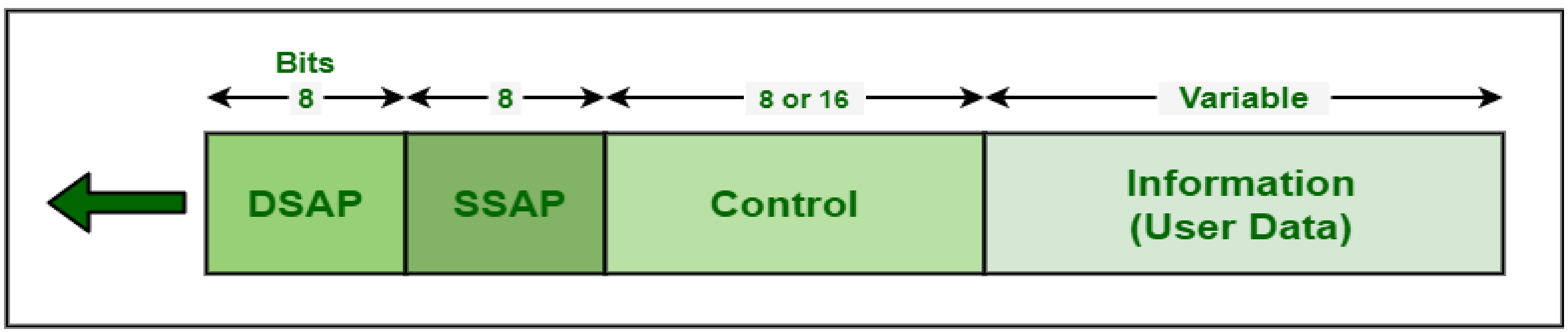


PDU Format

Medium Access Control(MAC)

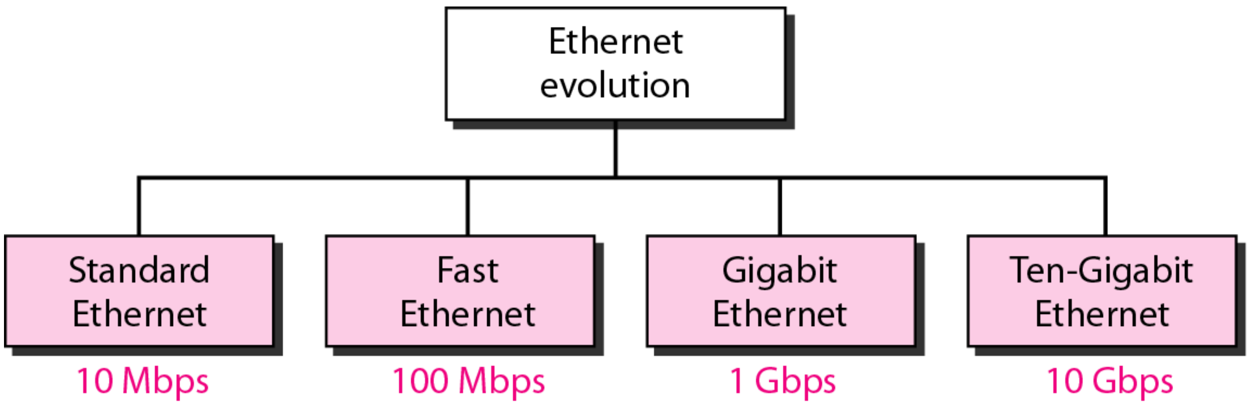
- IEEE Project 802 has created a sub-layer called MAC that defines specific access methods for each LAN.
- In contrast to LLC sub-layer, the MAC sub layer contains a number of distinct modules; each defines the access method and the framing format specific to the LAN protocol.

Physical layer

- The physical layer is dependent on the implementation and the type of physical media used.

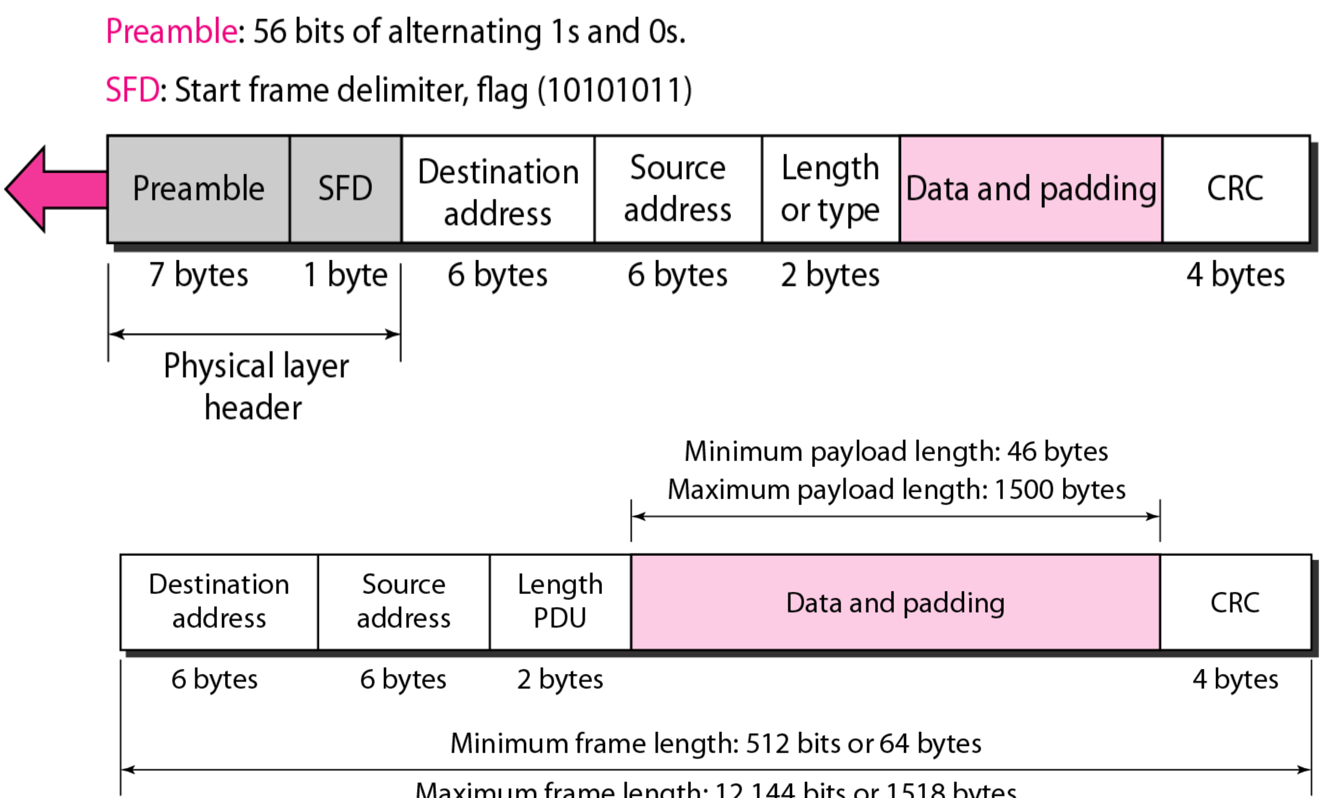
Standard Ethernet

- The original Ethernet was created in 1976 at Xerox’s Palo Alto Research Center (PARC). Since then, it has gone through four generations



MAC Sublayer

- Preamble: Alternate 0's and 1's that alerts the receiving system to the coming frame and enables it to synchronize its input timing



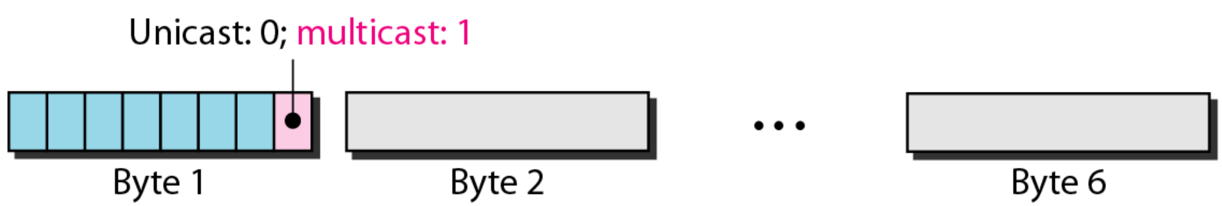
Addressing

- Ethernet address in hexadecimal notation

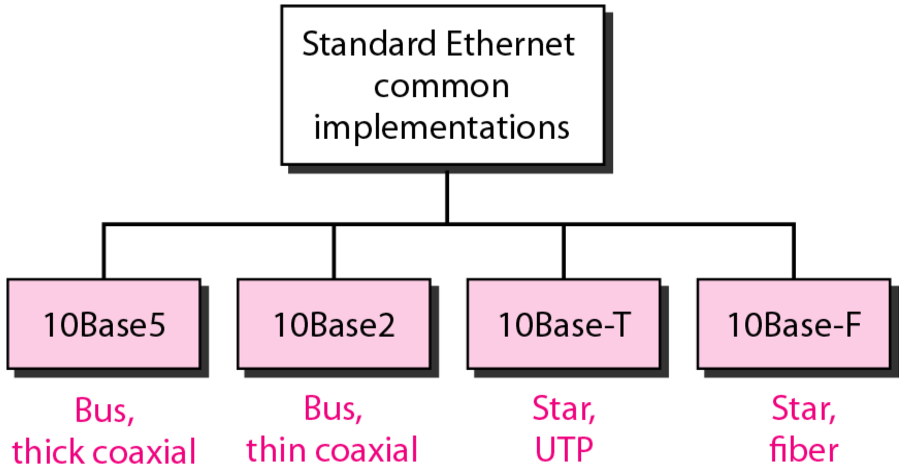
06 : 01 : 02 : 01 : 2C : 4B

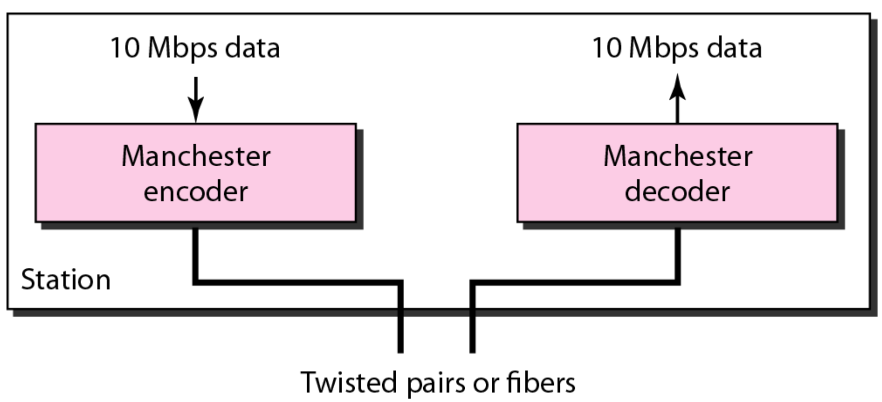
6 bytes = 12 hex digits = 48 bits

- The least significant bit of the first byte defines the type of address.
  If the bit is 0, the address is unicast; otherwise, it is multicast
- The broadcast destination address is a special case of the multicast address in which all bits are 1s



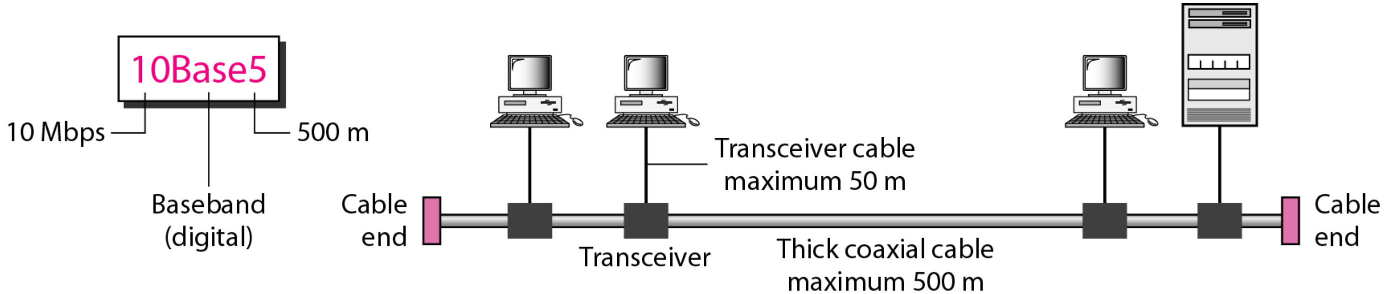
Physical Layer: Ethernet



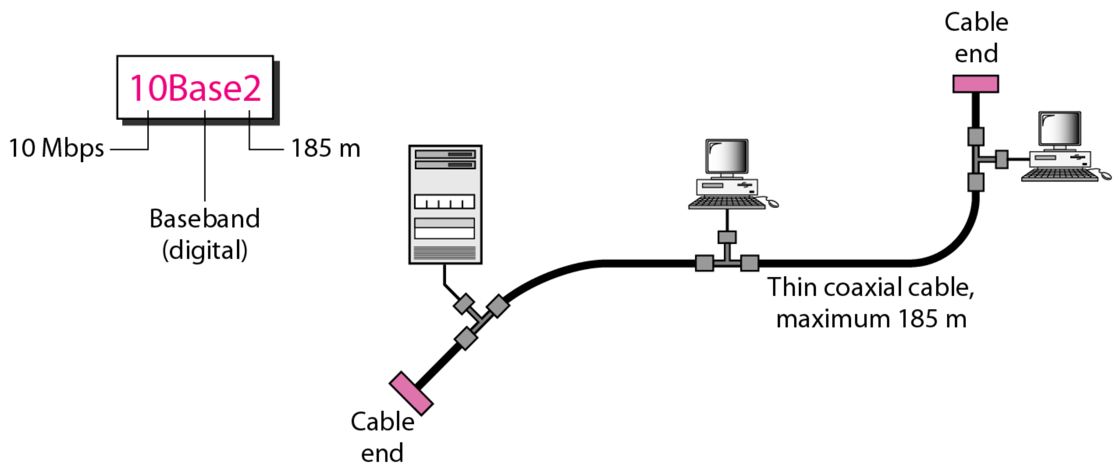


- This type of network involves using switches or hubs to improve network performance.
- Each workstation in this network has its own dedicated connection, which improves the speed and efficiency of data transfer.
- Switch Ethernet supports a wide range of speeds, from 10 Mbps to 10 Gbps, depending on the version of Ethernet being used.

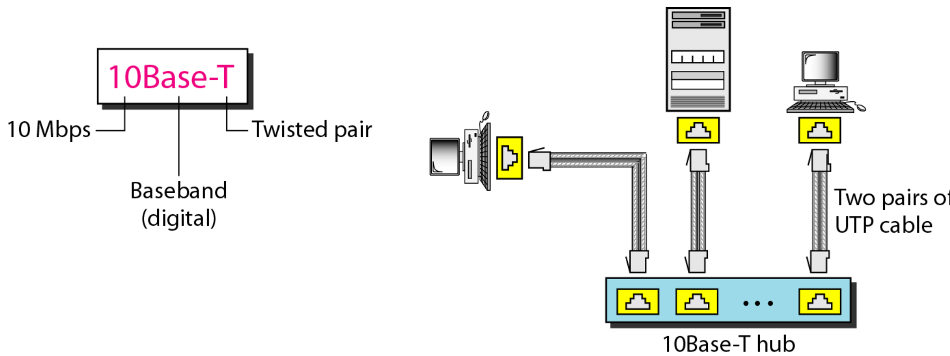
10Base5: Thick Ethernet



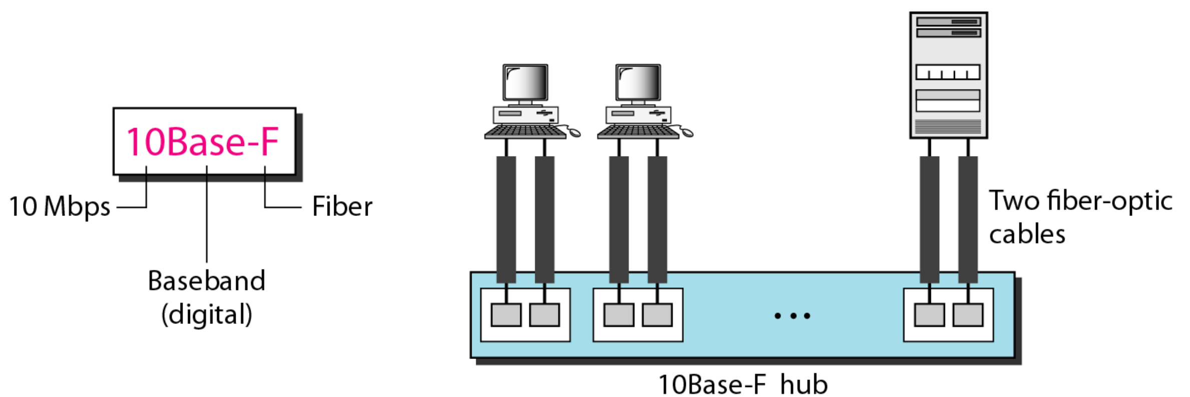
10Base2: Thin Ethernet



10BaseT: Twisted-Pair Ethernet



10Base-F: Fiber Ethernet



Summary of Standard Ethernet

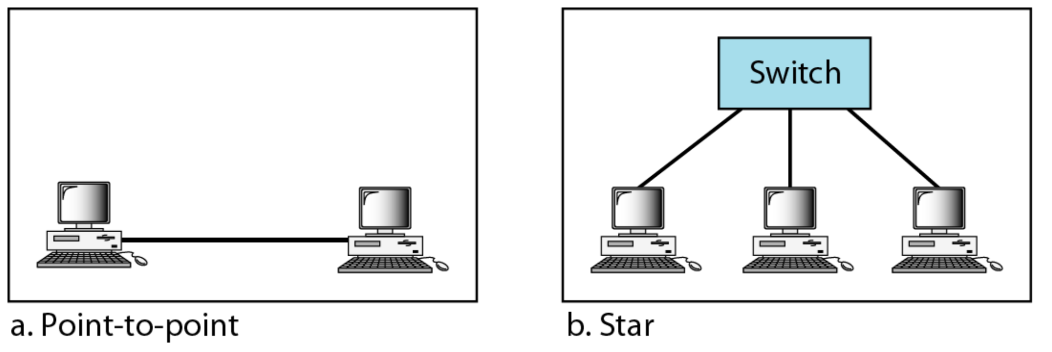
Characteristics	*10Base5*	*10Base2*	*10Base-T*	*10Base-F*
Media	Thick coaxial cable	Thin coaxial cable	2 UTP	2 Fiber
Maximum length	500 m	185 m	100 m	2000 m
Line encoding	Manchester	Manchester	Manchester	Manchester

Fast Ethernet

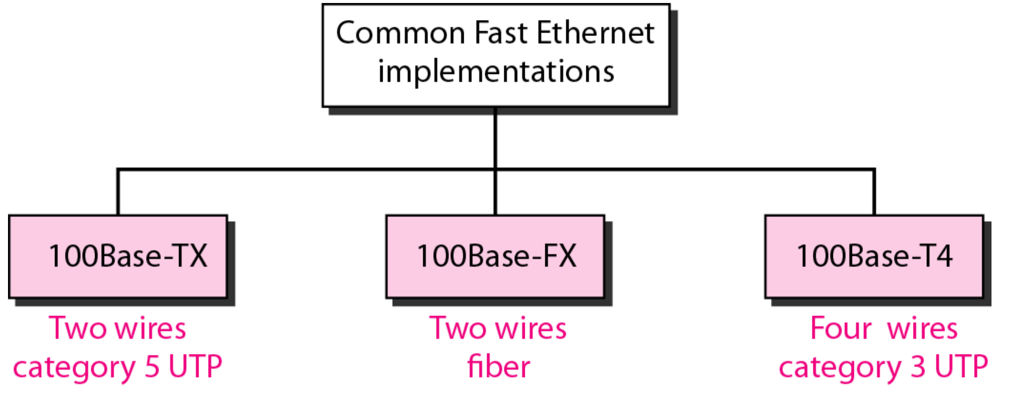
- This type of Ethernet network uses cables called twisted pair or CAT5.
- It can transfer data at a speed of around 100 Mbps (megabits per second).
- Fast Ethernet uses both fiber optic and twisted pair cables to enable communication.

Fast Ethernet: Physical Layer

- Topology



- Implementation

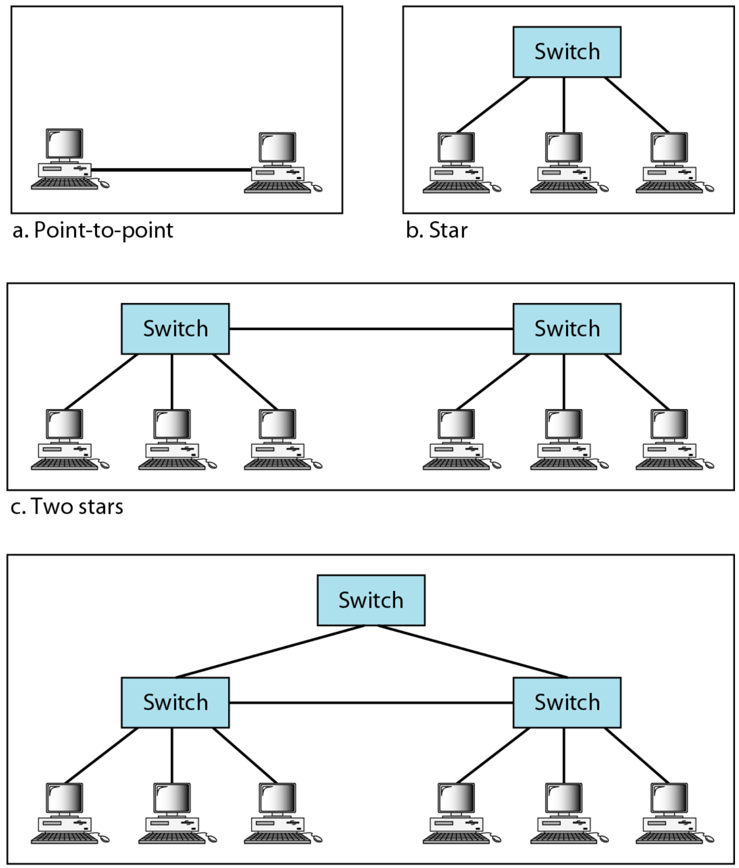


Gigabit Ethernet

- This is an upgrade from Fast Ethernet and is more common nowadays.
- It can transfer data at a speed of 1000 Mbps or 1 Gbps (gigabit per second).
- Gigabit Ethernet also uses fiber optic and twisted pair cables for communication.
- It often uses advanced cables like CAT5e, which can transfer data at a speed of 10 Gbps.

Gigabit Ethernet: Physical Layer

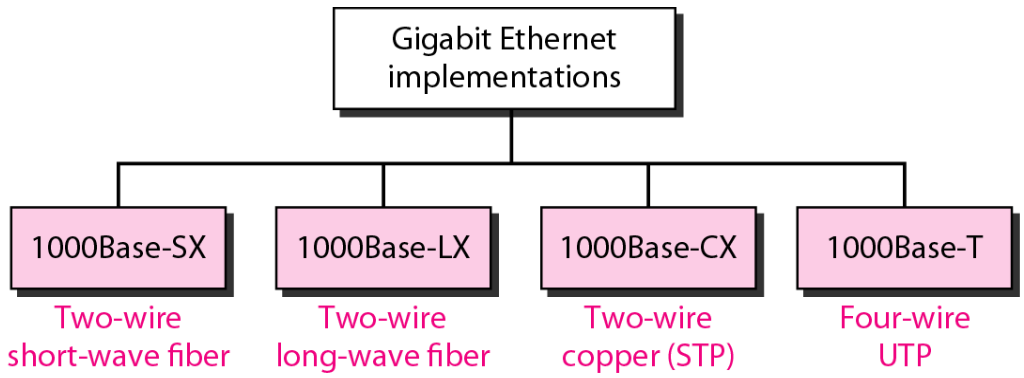
- Topology



a. Point-to-point
b. Star
c. Two stars
d. Hierarchy of stars

Gigabit Ethernet: Physical Layer

- Implementation



10-Gigabit Ethernet

- This is an advanced and high-speed network that can transmit data at a speed of 10 gigabits per second.
- It uses special cables like CAT6a or CAT7 twisted-pair cables and fiber optic cables.
- With the help of fiber optic cables, this network can cover longer distances, up to around 10,000 meters.

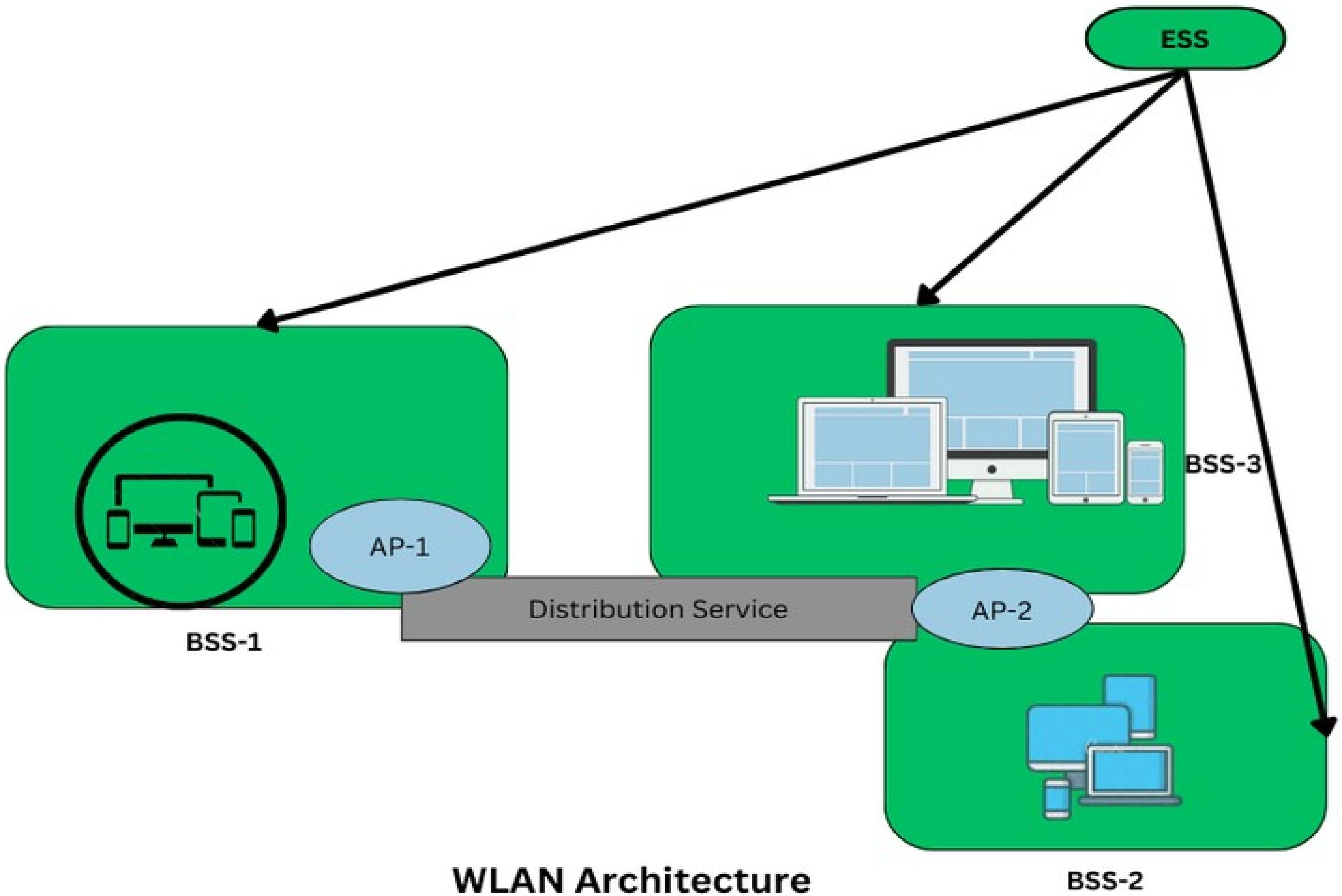
Wireless
Local Area Networks

Wireless Local Area Networks

- WLAN is a local area network that uses radio communication to provide mobility to the network users while maintaining the connectivity to the wired network.
- A WLAN basically, extends a wired local area network.
- WLAN's are built by attaching a device called the access point(AP) to the edge of the wired network.
- Clients communicate with the AP using a wireless network adapter which is similar in function to an ethernet adapter.
- It is also called a LAWN is a Local area wireless network.

- The performance of WLAN is high compared to other wireless networks.
- The standards of WLAN are HiperLAN, Wi-Fi, and IEEE 802.11.
- It offers service to the desktop laptop, mobile application, and all the devices that work on the Internet.
- WLAN is an affordable method and can be set up in 24 hours.
- Most latest brands are based on IEE 802.11 standards, which are the WI-FI brand name.

WLAN Architecture



- **Stations:** Stations consist of all the equipment that is used to connect all wireless LANs. Each station has a wireless network controller.
- **Base Service Set(BSS):** It is a group of stations communicating at the physical layer.
- **Extended Service Set(ESS):** It is a group of connected Base Service Set(BSS).
- **Distribution Service (DS):** It connects all Extended Service Set(ESS).

Types of WLANs

- **Infrastructure:** In Infrastructure mode, all the endpoints are connected to a base station and communicate through that; and this can also enable internet access.
- A WLAN infrastructure can be set up with: a wireless router (base station) and an endpoint (computer, mobile phone, etc).
- An office or home WiFi connection is an example of Infrastructure mode.

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Ad Hoc:

•In Ad Hoc mode WLAN connects devices without a base station, like a computer workstation.

•An Ad Hoc WLAN is easy to set up it provides peer-to-peer communication.

• It requires two or more endpoints with built-in radio transmission

Wireless LANs

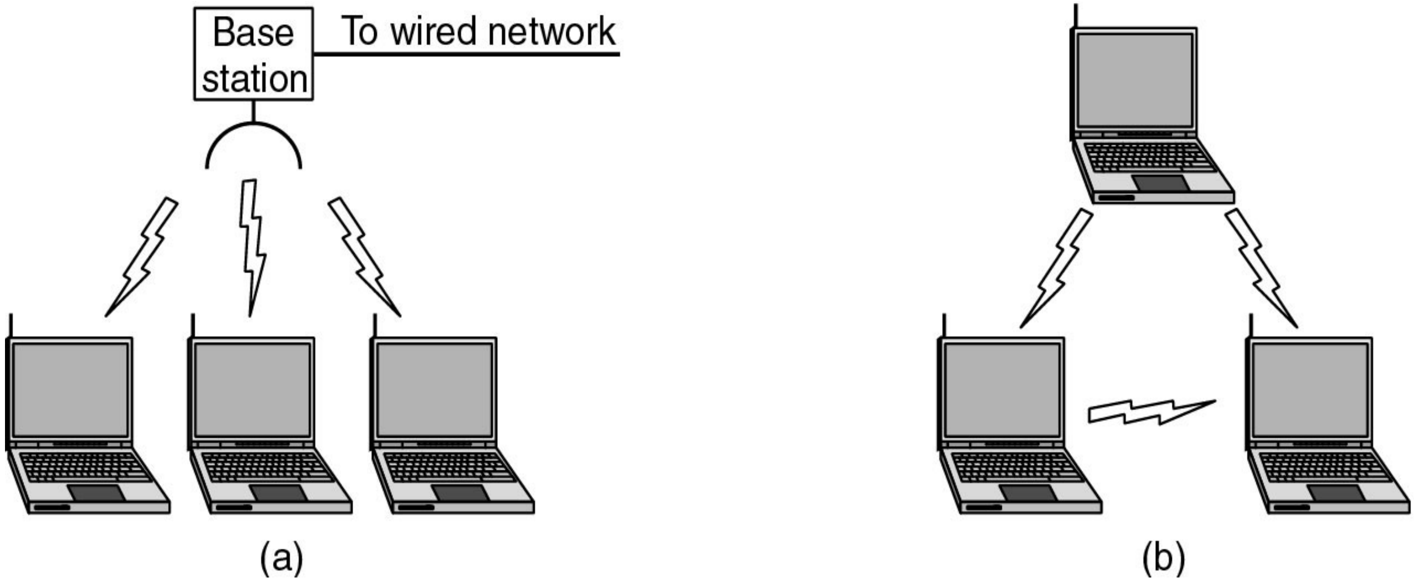


Figure 1-36.(a) Wireless networking with a base station. (b) Ad hoc networking.

WLAN Access Protocols

- These protocols define how devices manage access to the shared wireless medium:
- **CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance):**

The primary mechanism in Wi-Fi, where devices "listen" to the channel before transmitting to avoid data collisions.

Advantages of WLAN

- Installation speed and simplicity.
- Installation flexibility.
- Reduced cost of ownership.
- Reliability.
- Mobility.
- Robustness.

Disadvantages of WLAN

- Slower bandwidth.
- Security for wireless LANs is the prime concern.
- Less capacity.
- Wireless networks cost four times more than wired network cards.

Types of Bridges

•There are three types of bridges in computer networks, which are as follows:

1.Transparent Bridge:

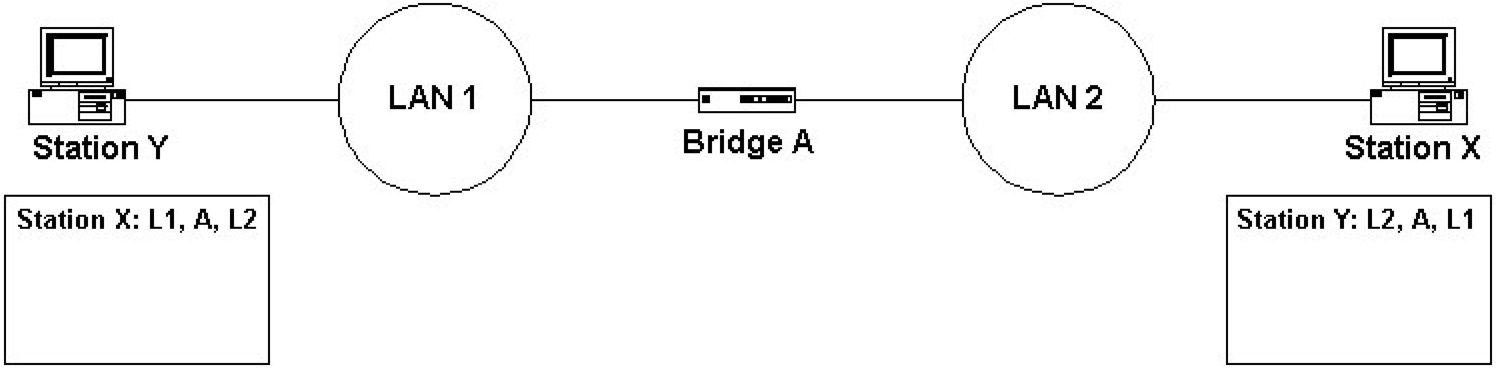
•Transparent bridges are invisible to other devices on the network. This bridge doesn't reconfigure the network on the addition or deletion of any station.

•The prime function of the transparent bridge is to block or forward the data according to the MAC address.

2.Source Routing Bridge:

•Source routing bridges were developed and designed by IBM specifically for token ring networks.

•The frame's entire route is embedded with the data frames by the source station to perform the routing operation so that once the frame is forwarded it must follow a specific defined path/route.

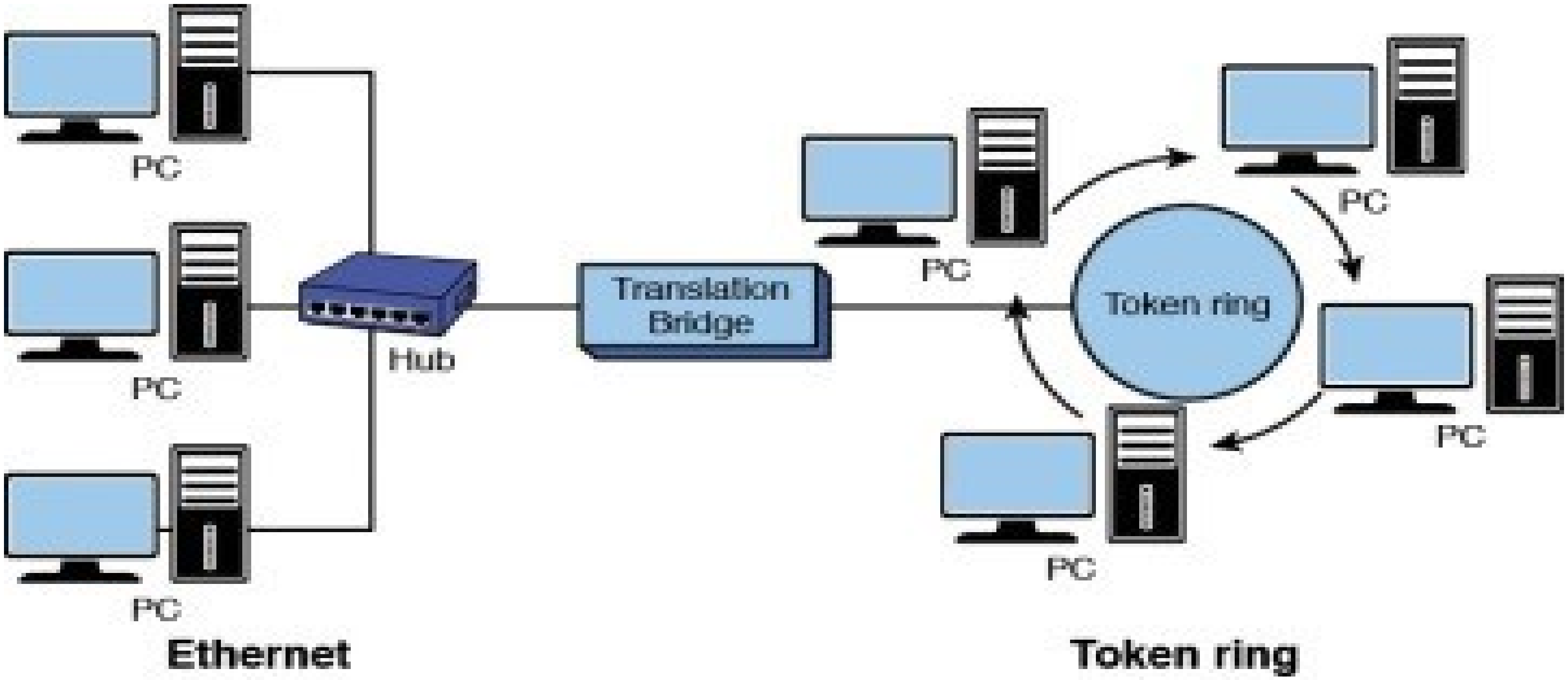

LAN 1
LAN 2
Bridge A
Station Y
Station X
Station X: L1, A, L2
Station Y: L2, A, L1

Translational Bridge:

•Translational bridges convert the received data from one networking system to another.

• It is used to communicate or transmit data between two different types of networking systems.

•Like if we are sending data from a token ring to an Ethernet cable, the translational cable will be used to connect both the networking system and transmit data



Translation bridge connecting different N/ws

Advantages

- Bridges can be used as a network extension like they can connect two network topologies together.
- Highly reliable and maintainable.
- Simple installation, no requirement of any extra hardware or software except the bridge itself.

- **Disadvantages**
- Expensive as compared to hubs and repeaters.
- Slow in speed.
- Poor performance as additional processing is required to view the MAC address of the device on the network.